

GR-01 : FUNCTIONS

SINGLE CORRECT

Q.(1)

Which of the following function is surjective but not injective

(A) $f: \mathbb{R} \rightarrow \mathbb{R} \quad f(x) = x^4 + 2x^3 - x^2 + 1$

(B) $f: \mathbb{R} \rightarrow \mathbb{R} \quad f(x) = x^3 + x + 1$

(C) $f: \mathbb{R} \rightarrow \mathbb{R}^+ \quad f(x) = \sqrt{1+x^2}$

(D) $f: \mathbb{R} \rightarrow \mathbb{R} \quad f(x) = x^3 + 2x^2 - x + 1$

Q.(2)

Let $f(x) = \max.\{\sin t : 0 \leq t \leq x\}$

$g(x) = \min.\{\sin t : 0 \leq t \leq x\}$

and $h(x) = [f(x) - g(x)]$

where $[]$ denotes greatest integer function, then the range of $h(x)$ is-

(A) $\{0,1\}$

(B) $\{1,2\}$

(C) $\{0,1,2\}$

(D) $\{-3,-2,-1,0,1,2,3\}$

MULTIPLE CORRECT

Q.(3)

Let $f: \left(-\frac{\pi}{2}, a\right] \rightarrow (-\infty, b]$ be an invertible function with longest domain defined as

$f(x) = \tan x + \cot x - 2$ then -

(A) Ordered pair (a,b) is $\left(-\frac{\pi}{3}, -\frac{(4+2\sqrt{3})}{\sqrt{3}}\right)$

(B) Ordered pair (a,b) is $\left(-\frac{\pi}{4}, -4\right)$

(C) $f^{-1}(x) = \tan^{-1}\left(\frac{x+2+\sqrt{x^2+4x}}{2}\right)$

(D) $f^{-1}(x) = \tan^{-1}\left(\frac{x+2-\sqrt{x^2+4x}}{2}\right)$

Q.(4)

Which of the following statement(s) is(are) correct ?

- (A) If f is a one-one mapping from set A to A , then f is onto.
- (B) If f is an onto mapping from set A to A , then f is one-one
- (C) Let f and g be two functions defined from $\mathbb{R} \rightarrow \mathbb{R}$ such that $g \circ f$ is injective, then f must be injective.
- (D) If set A contains 3 elements while set B contains 2 elements, then total number of functions from A to B is 8.

Q.(5)

If $f(x) = x^3 + x^2 + 2x$, $g(x) = e^x$ (where $g(x)$ is invertible function), then which of the following statement(s) is/are correct ?

- (A) $g \circ f(x) \leq g(x)$ for $x \in (-\infty, 0]$
- (B) $g^{-1} \circ f(x) \geq g^{-1}(x)$ for $x \in (0, \infty)$
- (C) $g^{-1} \circ f(x) \leq g^{-1}(x)$ for $x \in (0, \infty)$
- (D) $g^{-1} \circ f(x) \geq g^{-1}(x)$ for $x \in (-\infty, 0]$

Q.(6) Let $f: \mathbb{A} \rightarrow \mathbb{B}$ and $g: \mathbb{C} \rightarrow \mathbb{A}$ be functions. Then which of the following options is/are correct?

- (A) If f and g are Injective, then $f \circ g$ is also Injective.
- (B) If f and g are Surjective, then $f \circ g$ is also Surjective.
- (C) If $f \circ g$ is Injective, then g MUST be Injective.
- (D) If $f \circ g$ is Injective, then f need not be Injective.

Q.(7) Let $f: \mathbb{A} \rightarrow \mathbb{B}$ and $g: \mathbb{C} \rightarrow \mathbb{A}$ be functions. Then which of the following options is/are correct?

- (A) If $f \circ g$ is Surjective, then f MUST also be Surjective.
- (B) If $f \circ g$ is Surjective, then g need not be Surjective.
- (C) If $f \circ g$ is Injective and g is Surjective, then f MUST be Injective.
- (D) If f and g are Bijective, then $f \circ g$ need not be Bijective.

Q.(8) Consider the following sets

$$\mathbb{A} = \left\{ \frac{x^2 - 2x + 4}{x^2 + 2x + 4} \in \mathbb{R} : x \in \mathbb{R} \right\},$$

$$\mathbb{B} = \left\{ \frac{x^2 - 2x - 4}{x^2 + 2x - 4} \in \mathbb{R} : x \in \mathbb{R} \right\}.$$

Then which of the following options is/are correct?

(A) $\left\{1, \frac{3}{7}, \frac{7}{3}, 3, \frac{1}{3}, \frac{7}{19}, \frac{19}{7}\right\} \subseteq \mathbb{A}$

(B) $\{-13, 2301, \pi, -e\} \subseteq \mathbb{B}$

(C) $\left\{1, \frac{3}{7}, \frac{7}{3}, 3, \frac{1}{3}, \frac{19}{39}, \frac{39}{19}\right\} \subseteq \mathbb{A}$

(D) $\{-1301, 23, -\pi, e\} \subseteq \mathbb{B}$

Q.(9) Which of the following options is/are correct, if

$$f(x) = \max(x^2, |2 - x|) \text{ and } g(x) = \min(x^3, 3x - 2)?$$

(A) If $x \in (-\infty, -2]$, then $f(g(x)) = x^6$

(B) If $x \in (-2, 0]$, then $f(g(x)) = 9x^2 - 12x + 4$

(C) If $x \in (0, 1]$, then $f(g(x)) = 4 - 3x$

(D) If $x \in (-2, 1]$, then $g(f(x)) = 4 - 3x$

Q.(10) Let $a, b, c \in \mathbb{R}$, $a + b + c = 3$ and functions f, g, h be such that

$$f: \mathbb{R} \rightarrow \left[-\frac{1}{2}, \frac{1}{2}\right]; f(x) = \frac{x}{1 + x^2},$$

$$g(x) = ax^2 + bx + c; g(x + y) = g(x) + g(y) + xy \forall x, y \in \mathbb{R} \text{ and}$$

$$h(x) + 2h\left(\frac{1}{x}\right) = 3x \forall x \in \mathbb{R} - \{0\}.$$

If $|\mathbb{X}|$ denotes the cardinal number of set $\mathbb{X}$, then which of the following options is/are correct?

(A) f is Injective function.

(B) f is Surjective function.

(C) $g(1) + g(2) + g(3) + \dots + g(10) = 330$

(D) $|\{x \in \mathbb{R} : h(x) = h(-x)\}| = 2.$

NUMERICAL TYPE

Q.(11)

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function satisfying $f(10 - x) = f(x)$ and $f(2 - x) = f(2 + x) \forall x \in \mathbb{R}$. If $f(0) = \lambda$ (where $\lambda \in \mathbb{R}$). Then the minimum possible number of values of x satisfying $f(x) = \lambda$ in the interval $[0, 38]$, is

Q.(12)

Let $A = \{1, 2, 3, 4, 5\}$ & $B = \{a, b, c, d\}$. A function is defined from A to B . 'p' be the number of one-one functions from A to B and 'q' be the number of onto function from A to B . Then $(p + q)$ equals

Q.(13) Let

$$f: \mathbb{A} \rightarrow \left[-\frac{1}{2}, 0\right]; f^3(x) - 3x f(x) + x^3 + 1 = 0.$$

If f is an invertible function, then number of solutions of

$$(f(f(x)))^2 = \frac{3}{2}f^{-1}(x) + 1$$

is equal to

Q.(14) Let $[x] = \max\{a \in \mathbb{Z} : a \leq x\}$, $\{x\} = x - [x]$,

$$\mathbb{A} = \left\{x \in \mathbb{R} : \frac{[x+1]^2}{6} = \frac{x}{3} + \frac{\{x\}}{2}\right\},$$

$$\mathbb{B} = \left\{x \in \mathbb{R}^+ : \{x^2\} = \left\{\frac{1}{x}\right\}, x^2 \in (2, 3)\right\}.$$

If $|\mathbb{X}|$ denotes the cardinal number of set $\mathbb{X}$, then $|\mathbb{A}| + |\mathbb{B}|$ is equal to

Q.(15)

If $1 < x^2 < 9$, then number of solutions of the equation $\{[x]\} = \left\{\frac{1}{x^2}\right\}$ (where $\{.\}$ denote fractional part function) is -

Q.(16) Let $|\mathbb{X}|$ denotes the cardinal number of set $\mathbb{X}$,

$$\mathbb{A} = \{x \in \mathbb{R} : \sqrt{ax^2 + bx} \in \mathbb{R}\},$$

$$\mathbb{B} = \{\sqrt{ax^2 + bx} \in \mathbb{R} : x \in \mathbb{R}\},$$

$$\mathbb{C} = \{a \in \mathbb{R} : \mathbb{A} = \mathbb{B} \text{ for at least one } b \in \mathbb{R}^+\}.$$

Then absolute value of sum of elements of set $\mathbb{C}$ is equal to

Q.(17) Number of elements in the set

$$\{5x \in \mathbb{Z} : \log_3(\log_3|\cos x| - \log_{x^2-4x+6}|\cos x|) \in \mathbb{R}\}$$

is equal to

PASSAGES

Passage:

The function $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined as $f(x) = \frac{e^x - e^{-x}}{2}$ and function $g : \mathbb{R} \rightarrow [1, \infty)$ is defined as

$$g(x) = \frac{e^x + e^{-x}}{2}.$$

On the basis of above information, answer the following questions :

Q.(18)

Let $h(x) = \frac{f(x)}{g(x)}$, then $\frac{2h(x)}{1+h^2(x)}$ is equal to

- (A) 1 (B) 2 (C) $h(2x)$ (D) $h(x)f(x)$

Q.(19)

Number of integers contained in the range of the function $k(x) = \left[\frac{f(x)}{g(x)} \right]$ (where $[.]$ denotes G.I.F.) is

- (A) 0 (B) 1 (C) 2 (D) more than 2

MATRIX MATCHING TYPE

Q.(20)

Column-I

Column-II

(A) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as

(P) Odd

$$f(x) = e^{\operatorname{sgn} x} + e^{x^2}, \text{ where } \operatorname{sgn} x$$

denotes signum function of x , then $f(x)$ is

(B) Let $f : (-1, 1) \rightarrow \mathbb{R}$ defined as

(Q) Even

$$f(x) = x[x^4] + \frac{1}{\sqrt{1-x^2}}$$

where $[x]$ denotes greatest integer less than or equal to x , then $f(x)$ is

(C) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as

(R) Neither odd nor even

$$f(x) = \frac{x(x+1)(x^4+1) + 2x^4 + x^2 + 2}{x^2 + x + 1} \text{ then } f(x) \text{ is}$$

(D) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as

(S) One-one

$$f(x) = x + 3x^3 + 5x^5 + \dots + 101x^{101}$$

then $f(x)$ is

(T) Many-One

KEY

1. D	2. C	3. BD	4. CD	5. AB
6. ABCD	7. ABC	8. ABCD	9. ACD	10. BCD
11. 13	12. 240	13. 1	14. 4	15. 4
16. 4	17. 9	18. C	19. C	20. A-RT, B-QT, C-RT, D-PS

GR-02 : LIMITS AND CONTINUITY

SINGLE CORRECT

Q.(1)

The value of $\lim_{x \rightarrow 0} \frac{\ln(\sec(ex) \sec(e^2x) \dots \sec(e^{50}x))}{e^2 - e^{2\cos x}}$ is equal to

- (A) $\frac{e^{100} - 1}{(e^2 - 1)}$ (B) $\frac{e^{100} - 1}{2(e^2 - 1)}$ (C) $\frac{2(e^{50} - 1)}{e^2 - 1}$ (D) $\frac{e^2(e^{100} - 1)}{2(e^2 - 1)}$

Q.(2)

$\lim_{x \rightarrow 0} \frac{\sin x^4 - x^4 \cos x^4 + x^{20}}{x^4(e^{2x^4} - 1 - 2x^4)}$ is equal to

- (A) 0 (B) $-\frac{1}{6}$ (C) $\frac{1}{6}$ (D) does not exist

Q.(3)

Let $g(x) = \tan^{-1} |x| - \cot^{-1} |x|$, $f(x) = \frac{[x]}{[x+1]} \{x\}$, $h(x) = |g(f(x))|$ where $\{x\}$ denotes fractional part and

$[x]$ denotes the integral part then which of the following holds good ?

- (A) h is continuous at $x = 0$ (B) h is discontinuous at $x = 0$
(C) $h(0^-) = \pi/2$ (D) $h(0^+) = -\pi/2$

Q.(4) Let f be a positive continuous function satisfying

$$f(x^3 - 4x) = \frac{2}{f(2^x + 7) - 1} \forall x \in \mathbb{R}.$$

Also, let g be a continuous function satisfying

$$g(x) = g(2x) \forall x \in \mathbb{R}, g(1) = \frac{\pi}{2},$$

then find

$$\lambda = f(15) + \lim_{x \rightarrow 0} \frac{\cos^2 g(x) + 1 - \sin^3 g(x)}{\sin^2 x}.$$

(A) 1

(B) 2

(C) 3

(D) 4

Q.(5) Which of the following options is/are correct, if

$$\mathbb{S} : \text{If } a_{n-1} = \frac{a_n - n}{n} \forall n \in \mathbb{N} - \{1\}, a_1 = 1, \text{ then } \lim_{n \rightarrow \infty} \prod_{r=1}^n \left(1 + \frac{1}{a_r}\right) = e.$$

$$\mathbb{T} : \lim_{n \rightarrow \infty} \left(\frac{n^2 + n + 1}{n + 1} - (an + b) \right) = 4 \Rightarrow 7a - 2b = 15.$$

 (A) $\mathbb{S}$ is True and $\mathbb{T}$ is False

 (B) $\mathbb{S}$ is False and $\mathbb{T}$ is True

 (C) Both $\mathbb{S}$ and $\mathbb{T}$ are True

 (D) Both $\mathbb{S}$ and $\mathbb{T}$ are False.

MULTIPLE CORRECT

Q.(6)

Which of the following limits are non existent ?

(A) $\lim_{x \rightarrow 0} \frac{1}{x} \sqrt{\frac{1 - \cos x}{1 + \cos x}}$

(B) $\lim_{x \rightarrow 0} \frac{|x|}{x}$

(C) $\lim_{x \rightarrow 0} \frac{\cot^{-1} \frac{1}{x}}{x}$

(D) $\lim_{x \rightarrow \pi/2} \cos^{-1} [\cot x]$ where $[\cdot]$ is greatest integer function

Q.(7)

Which of the following is/are correct :

(A) $\lim_{x \rightarrow 0} \frac{\sin 4x}{1 - \sqrt{1-x}} = 8$

(B) $\lim_{x \rightarrow \pi/2} \frac{\cos 3x + 3 \cos x}{\left(\frac{\pi}{2} - x\right)^3} = 2$

(C) $\lim_{x \rightarrow \pi/6} \frac{2 - \sqrt{3} \cos x - \sin x}{(6x - \pi)^2} = \frac{1}{36}$

(D) $\lim_{x \rightarrow \pi} \frac{1 - \sin \frac{x}{2}}{\cos \frac{x}{2} \left(\cos \frac{x}{4} - \sin \frac{x}{4} \right)} = \frac{1}{2}$

Q.(8)

If $L = \lim_{x \rightarrow 1} \frac{(1-x)(1-x^2)(1-x^3) \dots (1-x^{2n})}{[(1-x)(1-x^2)(1-x^3) \dots (1-x^n)]^2}$ then L can be equal to

(A) $\prod_{r=1}^n \frac{n+r}{r}$

(B) $\frac{1}{n!} \prod_{r=1}^n (4r-2)$

(C) The sum of the coefficients of two middle terms in the expansion of $(1+x)^{2n-1}$.

(D) The coefficient of x^n in the expansion of $(1+x)^{2n}$.

Q.(9)

Let $f(x) = \frac{\cos^{-1}(\cos x)}{x}$, then which of the following is(are) correct (where $[.]$ denotes greatest integer function) ?

(A) $\lim_{x \rightarrow 0^+} [f(x)] = 1$

(B) $\lim_{x \rightarrow 0^-} [f(x)] = 1$

(C) $\lim_{x \rightarrow 0^+} [f(x)] = 0$

(D) $\lim_{x \rightarrow 0^-} [f(x)] = -1$

Q.(10)

Let $f(x) = \begin{cases} |x|, & -2 \leq x \leq 0 \\ e^x - 1, & 0 < x \leq 5 \end{cases}$, $g(x) = \begin{cases} -x - 1, & -1 \leq x \leq 0 \\ \frac{4x-3}{3}, & 0 < x \leq 3 \end{cases}$, then -

(A) $\lim_{x \rightarrow 0} f(g(x)) = 1$

(B) $\lim_{x \rightarrow 0^-} f(g(2^x)) = 1$

(C) $\lim_{x \rightarrow 0^+} f(g(2^x)) = e^{1/3} - 1$

(D) $\lim_{x \rightarrow 0^+} f(g(2^x - 1)) = \lim_{x \rightarrow 0^-} f(g(2^x - 1))$

Q.(11) Let $\lfloor x \rfloor = \max\{a \in \mathbb{Z} : a \leq x\}$, Then which of the following options is/are correct?

 (A) If f is a continuous function such that $\lfloor f(-1) \rfloor = 2$, $\lfloor f(0) \rfloor = -2$, then $f(x) = 0$ has at least one solution in $(-1, 0)$.

 (B) If f is a dis-continuous at $x = a$, and g is continuous at $x = a$, $g(a) = 0$, then $f(x)g(x)$ must be continuous at $x = a$.

 (C) If $f(x) + \sin x$ is continuous and $g(x) - e^x$ is dis-continuous at $x = a$, then $f(x)g(x)$ must be dis-continuous at $x = a$.

 (D) It is possible that $f(x)$ and $g(x) = \max\{t \in \mathbb{Z} : t \leq f(x)\}$ are both continuous at $x = a$, $f(a) \in \mathbb{Z}$.

Q.(12) Which of the following options is/are correct, if

$$f(x) = \lim_{n \rightarrow \infty} \frac{\sin \pi x - x^{2n} \sin(x-1)}{1 + x^{2n+1} - x^{2n}}; n \in \mathbb{N}, \text{ and}$$

$$\mathbb{A} = \{a \in \mathbb{Z} : g(x) = \lfloor x \rfloor^2 - \lfloor x^2 \rfloor \text{ is continuous at } x = a.\}$$

 Here $|\mathbb{X}|$ denotes the cardinal number of set $\mathbb{X}$

(A) $\lim_{x \rightarrow 1^+} f(x) = -1$

(B) $\lim_{x \rightarrow 1^-} f(x) = -1$

(C) f is dis-continuous at $x = 1$.

(D) $|\mathbb{A}| = 0$.

Q.(13) Let

$$\lambda = \lim_{x \rightarrow 0} \left(\frac{\tan x \cdot \cos \sin x - \sin \sin x}{\tan x \cdot \cos \tan x - \sin \tan x} \right), \mu = \lim_{x \rightarrow 0} \left(\frac{a - \sqrt{a^2 - x^2} - \frac{x^2}{4}}{x^4} \right); a > 0.$$

If μ is finite, then

- (A) $a - 2\lambda = 3$ (B) $64\mu + a = 3$ (C) $a - 2\lambda = 2$ (D) $32\mu = 1$

Q.(14) Let $\mu_1 = \lim_{a \rightarrow 0^+} (p), \mu_2 = \lim_{a \rightarrow 0^+} (q)$, where p, q are roots of

$$(\sqrt[3]{1+a} - 1)x^2 + (\sqrt{1+a} - 1)x + (\sqrt[6]{1+a} - 1) = 0; p > -1, \text{ and}$$

$$\lambda_1 = \lim_{x \rightarrow \infty} g(x), \lambda_2 = \lim_{x \rightarrow -\infty} g(x); \text{ where } g(x) = \frac{\ln(x^2 + e^x)}{\ln(x^4 + e^{2x})},$$

If $\mu_1 > \mu_2$, then which of the following options is/are correct?

- (A) $2\lambda_1 - \mu_2 = 2$ (B) $4\lambda_2 - 2\mu_1 = 3$ (C) $3\lambda_1 - \mu_1 = 2$ (D) $3\lambda_2 - 5\mu_2 = 3$

Q.(15) Let functions f and g are defined as

$$f(x) = \lim_{h \rightarrow 0} \frac{\sin(x+3h) - 3\sin(x+2h) + 3\sin(x+h) - \sin x}{h(\tan(x+2h) - 2\tan(x+h) + \tan x)} \text{ and}$$

$$g(x) = \frac{1 - x(1 + |1 - x|)}{|1 - x|} f\left(\frac{1}{1 - x}\right) \sin \frac{1}{1 - x} \sec^3 \frac{1}{1 - x}.$$

Then which of the following options is/are correct?

- (A) $\lim_{x \rightarrow 1^+} g(x)$ does not exist (B) $\lim_{x \rightarrow 1^-} g(x)$ does not exist
(C) $\lim_{x \rightarrow 1^+} g(x) = 0$ (D) $\lim_{x \rightarrow 1^-} g(x) = 0$

NUMERICAL TYPE

Q.(16)

If the value of limit $\lim_{x \rightarrow 0} \frac{(1+x)^{\frac{1}{x}} - e + \frac{ex}{2}}{x^2} = \frac{Ae}{B}$, where A and B are co-prime natural numbers, then find the value of (A+B) :-

Q.(17) Find the value of $(a - 2be + c) \in \mathbb{R}$, if $b, c \in \mathbb{R} - \{0\}$,

$$\lim_{x \rightarrow 0^+} \frac{(1-x)^{\frac{1}{x}} - \frac{1}{e}}{x^a} = b \text{ and } \lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{4\sqrt{2}(\sin x + \sin 3x)}{2 \sin 2x \sin \frac{3x}{2} + \cos \frac{5x}{2} - \sqrt{2} - \sqrt{2} \cos 2x - \cos \frac{3x}{2}} \right) = c.$$

Here, e denotes the base of natural logarithm.

Q.(18) If

$$f(x) = \begin{cases} \frac{a^{\sin x} - a^{\tan x}}{\tan x - \sin x}, & x > 0 \\ k, & x = 0 \\ \frac{\ln(1+x+x^2) + \ln(1-x+x^2)}{\sec x - \cos x}, & x < 0 \end{cases}$$

is continuous at $x = 0$, and

$$g(x) = \ln \left(2 - \frac{x}{a} \right) \cot(x-a); x \neq a, a > 0,$$

is continuous at $x = a$, then $\left| g\left(\frac{1}{e}\right) + 2e \right|$ is equal to (where $[x] = \max\{a \in \mathbb{Z} : a \leq x\}$)

Q.(19) Find the value of $\lambda + n$, where

$$\lim_{x \rightarrow 0} \left(\frac{\cos^2 x - \cos x - e^x \cos x + e^x - \frac{x^3}{2}}{x^n} \right) = \lambda (\in \mathbb{R} - \{0\}); n \in \mathbb{N}.$$

Q.(20) Let

$$\mathbb{A} = \left\{ a \in \mathbb{W} : \lim_{x \rightarrow 1} \left(\frac{-ax + \sin(x-1) + a}{x + \sin(x-1) - 1} \right)^{\frac{1-x}{1-\sqrt{x}}} = \frac{1}{4} \right\}, k = \lim_{x \rightarrow 0^+} \left(\frac{e^{x^x-1} - x^x}{(x^{2x} - 1)^2} \right).$$

Then $\max(\mathbb{A}) + 8k$ is equal to

PASSAGES
Passage-I

Let

$$f(x) = \begin{cases} \frac{e^{\frac{1}{x-1}} - 2}{e^{\frac{1}{x-1}} + 2}, & x \neq 1 \\ 1, & x = 1 \end{cases} \text{ and } g(x) = \begin{cases} a_n + \sin \pi x, & x \in [2n, 2n+1] \\ b_n + \cos \pi x, & x \in (2n-1, 2n) \end{cases}; n \in \mathbb{Z}.$$

Now, based on above information, answer the following questions

Q.(21) Which of the following options is/are correct?

$$(A) \lim_{x \rightarrow 1^+} f(x) = 1 \quad (B) \lim_{x \rightarrow 1^-} f(x) = -1 \quad (C) \lim_{x \rightarrow 1} f(x) = 1 \quad (D) \lim_{x \rightarrow 1} f(x) = -1$$

Q.(22) If g is continuous for all $x \in \mathbb{R}$, then

$$(A) a_{n-1} = b_{n-1} \quad (B) a_n - b_n = 1 \quad (C) a_n - b_{n+1} = 1 \quad (D) a_{n-1} - b_n = -1$$

Passage-II

 Let f and g are continuous functions defined from $[0,1] \rightarrow \mathbb{R}$ satisfying

$$\max\{f(x): x \in [0,1]\} = \max\{g(x): x \in [0,1]\}.$$

 Also $h(x) = x \cos(\pi x + \pi [x])$, where $[x] = \max\{a \in \mathbb{Z} : a \leq x\}$.

Now, based on above information, answer the following questions

Q.(23) Which of the following options is/are correct?

- (A) $(f(c))^2 + 3f(c) = (g(c))^2 + 3g(c)$ for some $c \in [0,1]$
 (B) $(f(c))^2 + f(c) = (g(c))^2 + 3g(c)$ for some $c \in [0,1]$
 (C) $(f(c))^2 + 3f(c) = (g(c))^2 + g(c)$ for some $c \in [0,1]$
 (D) $(f(c))^2 = (g(c))^2$ for some $c \in [0,1]$

Q.(24) The function h is dis-continuous at $x =$

- (A) -1 (B) 0 (C) 2 (D) 1

MATRIX MATCHING TYPE

Q.(25)

$$\text{Let } f(x) = \begin{cases} \frac{e^{(a+2)x} - 1}{x} & , x < 0 \\ c & , x = 0 \\ \frac{(x + b(x^4))^{1/2} - x^{1/2}}{b(x^{7/2})} & , x > 0 \end{cases}$$

is continuous function $\forall x \in \mathbb{R}$

Column-I

- (A) Complete set of value(s) of a is
 (B) Complete set of value(s) of b is
 (C) Complete set of value(s) of c is
 (D) Complete set of value(s) of $2ac$ is

Column-II

- (P) $\mathbb{R}^+$
 (Q) $\left\{-\frac{3}{2}\right\}$
 (R) $\left\{\frac{1}{2}\right\}$
 (S) $\mathbb{R} - \{0\}$
 (T) $\left\{-\frac{3}{2}, \frac{1}{2}\right\}$

KEY

1. B	2. C	3. A	4. B	5. C
6. ABCD	7. AC	8. ABCD	9. AD	10. ACD
11. AD	12. AC	13. AB	14. ABC	15. AD
16. 35	17. 10	18. 2	19. 4.5	20. 1
21. AB	22. BD	23. AD	24. ACD	25. A-Q, B-P, C-R, D-Q

PHYSICAL CHEMISTRY
Score Advance : Guided Revision Plan - Question Bank- (01)
Topic : Chemical Equilibrium & Ionic Equilibrium
MM : 68
Time : 45 Min.
SECTION-I : (i) Only One option correct Type

This section contains **8 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **3(-1)**

- 2.5 mL of $\frac{2}{5}$ M weak monoacidic base ($K_b = 1 \times 10^{-12}$ at 25°C) is titrated with $\frac{2}{15}$ M HCl in water at 25°C . The concentration of H^+ at equivalence point is
 ($K_w = 1 \times 10^{-14}$ at 25°C)
 (A) 3.7×10^{-13} M (B) 3.2×10^{-7} M (C) 3.2×10^{-2} M (D) 2.7×10^{-2} M
- How many moles of NH_3 must be added to 2.0 litre of 0.80 M AgNO_3 in order to reduce the Ag^+ concentration to 5×10^{-8} M. K_f of $[\text{Ag}(\text{NH}_3)_2]^+ = 10^8$
 (A) 0.4 (B) 2 (C) 3.52 (D) 4
- Let solubility of a sparingly soluble salt of strong base and weak acid in pure water is equal to s_1 , in a buffer solution with $\text{pH} < 7$ is equal to s_2 and in another buffer solution with $\text{pH} > 7$ is equal to s_3 . If hydrolysis is considered in all the solutions then the correct relation between s_1 , s_2 and s_3 is
 (A) $s_2 = s_2 = s_3$ (B) $s_1 > s_2 > s_3$ (C) $s_1 < s_2 < s_3$ (D) $s_2 > s_1 > s_3$
- For equilibrium at 300 K, $\text{LiCl} \cdot 3\text{NH}_3(\text{s}) \rightleftharpoons \text{LiCl} \cdot \text{NH}_3(\text{s}) + 2\text{NH}_3(\text{g})$; $K_p = 9 \text{ atm}^2$
 In 8.21 L vessel 1.4 mol of $\text{LiCl} \cdot \text{NH}_3(\text{s})$ and 2.8 moles of NH_3 are placed. Moles of $\text{LiCl} \cdot 3\text{NH}_3(\text{s})$ formed at equilibrium will be :
 (A) 1.4 (B) 0.7 (C) 0.9 (D) 1
- Two moles of an equimolar mixture of two alcohols $\text{R}_1\text{-OH}$ and $\text{R}_2\text{-OH}$ are esterified with one mole of acetic acid. If only 80% of the acid is consumed till equilibrium and the quantities of ester formed under equilibrium are in the ratio 3 : 2. What is the value of equilibrium constant for the esterification of $\text{R}_1\text{-OH}$?
 (A) 0.48 (B) 3.69 (C) 2.2 (D) 0.75
- An amount of 1 mole each of A and D is introduced in 1L container. Simultaneously the following two equilibria are established.
 $\text{A} \rightleftharpoons \text{B} + \text{C}; K_c = 10^6 \text{ M}$ and $\text{B} + \text{D} \rightleftharpoons \text{A}; K_c = 10^{-6} \text{ M}^{-1}$
 The equilibrium concentration of A will be
 (A) 10^{-6} M (B) 10^{-3} M (C) 10^{-12} M (D) 10^{-4} M
- Assuming no change in volume, calculate the minimum mass of NaCl necessary to dissolve 0.010 mol AgCl in 100 L solution. $[K_f(\text{AgCl}_2^-) = 3 \times 10^5, K_{sp}(\text{AgCl}) = 1 \times 10^{-10}]$
 (A) $\frac{10}{3}$ gm (B) 195 gm (C) 19.5 kg (D) 1950 gm
- The degree of hydrolysis of a salt of weak acid and weak base in its 0.1 M solution is found to be 50%. If the molarity of the solution is 0.2 M, the percentage hydrolysis of the salt should be
 (A) 100% (B) 50% (C) 25% (D) none of these

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(ii) One or more options correct Type

This section contains **5 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. **4(-1)**

9. Which of the following is true for alkaline aqueous solution?
- (A) $\text{pH} > \frac{\text{p}K_w}{2}$ (B) $\text{pH} > \text{pOH}$ (C) $\text{pOH} < \frac{\text{p}K_w}{2}$ (D) $\text{pH} < \text{pOH}$
10. An amount of 1 mole of $\text{PCl}_3(\text{g})$ and 1 mole of $\text{PCl}_5(\text{g})$ is taken in a vessel of 10 L capacity maintained at 400 K. At equilibrium, the moles of Cl_2 is found to be 0.004
- (A) K_C for the reaction ; $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$ is 0.0004 M
- (B) K_p for the reaction ; $\text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g}) \rightleftharpoons \text{PCl}_5(\text{g})$ is $0.0004 \times (0.0821 \times 400)$ atm.
- (C) If $\text{PCl}_3(\text{g})$ is added to the equilibrium mixture, ΔG at the new equilibrium becomes greater than the ΔG at old equilibrium
- (D) After equilibrium is achieved, moles of PCl_3 are doubled and moles of Cl_2 are halved simultaneously, then the partial pressure of PCl_5 remains unchanged.
11. At 90°C pure water has $[\text{H}^+] = 10^{-6}$ M. Based on this correct statement is (are) -
- (A) Increasing the temperature, increases degree of ionization of water.
- (B) At 60°C , pH of pure water will be less than 6
- (C) At 90°C , pH of a 10^{-6} M HCl solution will be less than 6
- (D) Ionization constant (K_a) of H_2O at 90°C is 10^{-12}
12. Select the correct statement(s) :
- (A) If pOH of a solution at 70°C is 7, then solution will be acidic
- (B) If $[\text{H}^+]$ concentration is decreases 100 times, pH of solution increases by 2 units
- (C) If solution of $\text{CH}_3\text{COONa}(\text{a})$ is diluted then pH increases.
- (D) Buffer capacity remains constant with dilution.
13. Aqueous solutions of HNO_3 , KOH , CH_3COOH and CH_3COONa of identical concentrations are provided. The pair(s) of solutions which form a buffer upon mixing is(are) -
- (A) HNO_3 and CH_3COOH (B) KOH and CH_3COONa
- (C) HNO_3 and CH_3COONa (D) CH_3COOH and CH_3COONa

SECTION-II: (Integer Second Decimal)

- This section contains 4 questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places; e.g. 6.25, 7.00, -0.33, -30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

						+	●
●	●	0	0	●	0		
1	1	1	1	1	1		
2	2	2	2	●	2		
3	3	3	3	3	3		
4	4	4	4	4	4		
5	5	5	5	5	●		
6	6	6	6	6	6		
7	7	●	●	7	7		
8	8	8	8	8	8		
9	9	9	9	9	9		

						●	-
●	●	●	0	0	●		
1	1	1	1	1	1		
2	2	2	2	●	2		
3	3	3	3	3	3		
4	4	4	4	4	4		
5	5	5	●	5	5		
6	6	6	6	6	6		
7	7	7	7	7	7		
8	8	8	8	8	8		
9	9	9	9	9	9		

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If ONLY the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- For the reaction : $A(g) \rightleftharpoons 2B(g)$, the equilibrium partial pressures of gases A and B is 4 bar and 2 bar, respectively. If the equilibrium mixture is slowly & isothermally compressed to 12 bar, maintaining equilibrium, then the new equilibrium pressure(in bar) of B(g) becomes.
- A weak base BOH was titrated against a strong acid. The pH at 1/4th equivalence point was 8.0. Enough strong base was now added (7.5 millimole) to completely convert the salt. The total volume was 10 ml. Find the pOH at this point.
- How much water (in L) should be added to 10 g CH_3COOH to give pOH equal to 11.
 $K_a(CH_3COOH) = 1.8 \times 10^{-5}$
- If $[H^+]$ of a 1 litre solution made by mixing 0.01 mol HCl with 1 mol BCl is xM, then the value of '1000x' is.

[Given : pK_b of BOH = 10, $\sqrt{5} = 2.236$]

SECTION-IV: (Matrix Match type)

This Section contains **1 question**. Question has **four statements** (A, B, C and D) given in **Column I** and five statements (P, Q, R, S and T) in **Column II**. Any given statement in Column I can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in Q and R, then for the particular question, against statement B, darken the bubbles corresponding to Q and R in the ORS.

8(0)

1. Column-I	Column-II
(A) $2AB_{3(g)} \rightleftharpoons A_{2(g)} + 3B_{2(g)}$	(P) Introduction of $B_{2(g)}$ at constant volume shift equilibrium in backward direction
(B) $AB_{3(g)} \rightleftharpoons AB_{(g)} + B_{2(g)}$	(Q) α increases on increasing temperature
(C) $2AB_{(g)} \rightleftharpoons A_{2(g)} + B_{2(g)}$	(R) On adding inert gas at constant pressure equilibrium concentration will change
(D) $AB_{2(s)} \rightleftharpoons A_{(g)} + B_{2(g)}$	(S) On increasing the pressure, concentration of all reactants increases
	(T) α is independent of equilibrium pressure.

ANSWER KEY**SECTION-I : (i) Only One option correct Type: 3(-1)**

- | | | | |
|------------|------------|-------------|-------------|
| 1. Ans.(D) | 2. Ans.(D) | 3. Ans.(D) | 4. Ans.(C) |
| 5. Ans.(B) | 6. Ans.(A) | 7. Ans. (C) | 8. Ans. (B) |

(ii) One or more options correct Type : 4(-1)

- | | | | |
|------------------|---------------|---------------|---------------|
| 9. Ans.(A, B, C) | 10. Ans.(A,D) | 11. Ans.(A,C) | 12. Ans.(A,B) |
| 13. Ans. (C, (D) | | | |

SECTION-II: (Integer Second Decimal) : 4(0)

- | | | |
|-------------------------|---------------|------------------------|
| 1. Ans.(3.00) | 2. Ans.(3.00) | 3. Ans. (2.94 to 3.00) |
| 4. Ans.(16.10 to 16.30) | | |

SECTION-IV : Matrix-Match Type : 8(0)

1. Ans.(A)-P,Q,R,S ; (B) P, Q, R, S, (C)- P, Q, R, S,T, (D)-P,Q

PHYSICAL CHEMISTRY
Score Advance : Guided Revision Plan - Question Bank- (01)
Topic : Chemical Equilibrium & Ionic Equilibrium
MM : 68
Time : 45 Min.
SECTION-I : (i) Only One option correct Type

This section contains **8 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **3(-1)**

- 2.5 mL of $\frac{2}{5}$ M weak monoacidic base ($K_b = 1 \times 10^{-12}$ at 25°C) is titrated with $\frac{2}{15}$ M HCl in water at 25°C . The concentration of H^+ at equivalence point is
 ($K_w = 1 \times 10^{-14}$ at 25°C)
 (A) 3.7×10^{-13} M (B) 3.2×10^{-7} M (C) 3.2×10^{-2} M (D) 2.7×10^{-2} M
- How many moles of NH_3 must be added to 2.0 litre of 0.80 M AgNO_3 in order to reduce the Ag^+ concentration to 5×10^{-8} M. K_f of $[\text{Ag}(\text{NH}_3)_2]^+ = 10^8$
 (A) 0.4 (B) 2 (C) 3.52 (D) 4
- Let solubility of a sparingly soluble salt of strong base and weak acid in pure water is equal to s_1 , in a buffer solution with $\text{pH} < 7$ is equal to s_2 and in another buffer solution with $\text{pH} > 7$ is equal to s_3 . If hydrolysis is considered in all the solutions then the correct relation between s_1 , s_2 and s_3 is
 (A) $s_2 = s_2 = s_3$ (B) $s_1 > s_2 > s_3$ (C) $s_1 < s_2 < s_3$ (D) $s_2 > s_1 > s_3$
- For equilibrium at 300 K, $\text{LiCl} \cdot 3\text{NH}_3(\text{s}) \rightleftharpoons \text{LiCl} \cdot \text{NH}_3(\text{s}) + 2\text{NH}_3(\text{g})$; $K_p = 9 \text{ atm}^2$
 In 8.21 L vessel 1.4 mol of $\text{LiCl} \cdot \text{NH}_3(\text{s})$ and 2.8 moles of NH_3 are placed. Moles of $\text{LiCl} \cdot 3\text{NH}_3(\text{s})$ formed at equilibrium will be :
 (A) 1.4 (B) 0.7 (C) 0.9 (D) 1
- Two moles of an equimolar mixture of two alcohols $\text{R}_1\text{-OH}$ and $\text{R}_2\text{-OH}$ are esterified with one mole of acetic acid. If only 80% of the acid is consumed till equilibrium and the quantities of ester formed under equilibrium are in the ratio 3 : 2. What is the value of equilibrium constant for the esterification of $\text{R}_1\text{-OH}$?
 (A) 0.48 (B) 3.69 (C) 2.2 (D) 0.75
- An amount of 1 mole each of A and D is introduced in 1L container. Simultaneously the following two equilibria are established.
 $\text{A} \rightleftharpoons \text{B} + \text{C}; K_c = 10^6 \text{ M}$ and $\text{B} + \text{D} \rightleftharpoons \text{A}; K_c = 10^{-6} \text{ M}^{-1}$
 The equilibrium concentration of A will be
 (A) 10^{-6} M (B) 10^{-3} M (C) 10^{-12} M (D) 10^{-4} M
- Assuming no change in volume, calculate the minimum mass of NaCl necessary to dissolve 0.010 mol AgCl in 100 L solution. $[K_f(\text{AgCl}_2^-) = 3 \times 10^5, K_{sp}(\text{AgCl}) = 1 \times 10^{-10}]$
 (A) $\frac{10}{3}$ gm (B) 195 gm (C) 19.5 kg (D) 1950 gm
- The degree of hydrolysis of a salt of weak acid and weak base in its 0.1 M solution is found to be 50%. If the molarity of the solution is 0.2 M, the percentage hydrolysis of the salt should be
 (A) 100% (B) 50% (C) 25% (D) none of these

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(ii) One or more options correct Type

This section contains **5 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. **4(-1)**

9. Which of the following is true for alkaline aqueous solution?
- (A) $\text{pH} > \frac{\text{p}K_w}{2}$ (B) $\text{pH} > \text{pOH}$ (C) $\text{pOH} < \frac{\text{p}K_w}{2}$ (D) $\text{pH} < \text{pOH}$
10. An amount of 1 mole of $\text{PCl}_3(\text{g})$ and 1 mole of $\text{PCl}_5(\text{g})$ is taken in a vessel of 10 L capacity maintained at 400 K. At equilibrium, the moles of Cl_2 is found to be 0.004
- (A) K_C for the reaction ; $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$ is 0.0004 M
- (B) K_p for the reaction ; $\text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g}) \rightleftharpoons \text{PCl}_5(\text{g})$ is $0.0004 \times (0.0821 \times 400)$ atm.
- (C) If $\text{PCl}_3(\text{g})$ is added to the equilibrium mixture, ΔG at the new equilibrium becomes greater than the ΔG at old equilibrium
- (D) After equilibrium is achieved, moles of PCl_3 are doubled and moles of Cl_2 are halved simultaneously, then the partial pressure of PCl_5 remains unchanged.
11. At 90°C pure water has $[\text{H}^+] = 10^{-6}$ M. Based on this correct statement is (are) -
- (A) Increasing the temperature, increases degree of ionization of water.
- (B) At 60°C , pH of pure water will be less than 6
- (C) At 90°C , pH of a 10^{-6} M HCl solution will be less than 6
- (D) Ionization constant (K_a) of H_2O at 90°C is 10^{-12}
12. Select the correct statement(s) :
- (A) If pOH of a solution at 70°C is 7, then solution will be acidic
- (B) If $[\text{H}^+]$ concentration is decreases 100 times, pH of solution increases by 2 units
- (C) If solution of $\text{CH}_3\text{COONa}(\text{a})$ is diluted then pH increases.
- (D) Buffer capacity remains constant with dilution.
13. Aqueous solutions of HNO_3 , KOH , CH_3COOH and CH_3COONa of identical concentrations are provided. The pair(s) of solutions which form a buffer upon mixing is(are) -
- (A) HNO_3 and CH_3COOH (B) KOH and CH_3COONa
- (C) HNO_3 and CH_3COONa (D) CH_3COOH and CH_3COONa

SECTION-II: (Integer Second Decimal)

- This section contains 4 questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places; e.g. 6.25, 7.00, -0.33, -30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

						+	●
●	●	0	0	●	0		
1	1	1	1	1	1		
2	2	2	2	●	2		
3	3	3	3	3	3		
4	4	4	4	4	4		
5	5	5	5	5	●		
6	6	6	6	6	6		
7	7	●	●	7	7		
8	8	8	8	8	8		
9	9	9	9	9	9		

						●	-
●	●	●	0	0	●		
1	1	1	1	1	1		
2	2	2	2	●	2		
3	3	3	3	3	3		
4	4	4	4	4	4		
5	5	5	●	5	5		
6	6	6	6	6	6		
7	7	7	7	7	7		
8	8	8	8	8	8		
9	9	9	9	9	9		

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If ONLY the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- For the reaction : $A(g) \rightleftharpoons 2B(g)$, the equilibrium partial pressures of gases A and B is 4 bar and 2 bar, respectively. If the equilibrium mixture is slowly & isothermally compressed to 12 bar, maintaining equilibrium, then the new equilibrium pressure(in bar) of B(g) becomes.
- A weak base BOH was titrated against a strong acid. The pH at 1/4th equivalence point was 8.0. Enough strong base was now added (7.5 millimole) to completely convert the salt. The total volume was 10 ml. Find the pOH at this point.
- How much water (in L) should be added to 10 g CH_3COOH to give pOH equal to 11.
 $K_a(CH_3COOH) = 1.8 \times 10^{-5}$
- If $[H^+]$ of a 1 litre solution made by mixing 0.01 mol HCl with 1 mol BCl is xM, then the value of '1000x' is.

[Given : pK_b of BOH = 10, $\sqrt{5} = 2.236$]

SECTION-IV: (Matrix Match type)

This Section contains **1 question**. Question has **four statements** (A, B, C and D) given in **Column I** and five statements (P, Q, R, S and T) in **Column II**. Any given statement in Column I can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in Q and R, then for the particular question, against statement B, darken the bubbles corresponding to Q and R in the ORS.

8(0)

1. Column-I	Column-II
(A) $2AB_{3(g)} \rightleftharpoons A_{2(g)} + 3B_{2(g)}$	(P) Introduction of $B_{2(g)}$ at constant volume shift equilibrium in backward direction
(B) $AB_{3(g)} \rightleftharpoons AB_{(g)} + B_{2(g)}$	(Q) α increases on increasing temperature
(C) $2AB_{(g)} \rightleftharpoons A_{2(g)} + B_{2(g)}$	(R) On adding inert gas at constant pressure equilibrium concentration will change
(D) $AB_{2(s)} \rightleftharpoons A_{(g)} + B_{2(g)}$	(S) On increasing the pressure, concentration of all reactants increases
	(T) α is independent of equilibrium pressure.

ANSWER KEY**SECTION-I : (i) Only One option correct Type: 3(-1)**

1. Ans.(D)	2. Ans.(D)	3. Ans.(D)	4. Ans.(C)
5. Ans.(B)	6. Ans.(A)	7. Ans. (C)	8. Ans. (B)

(ii) One or more options correct Type : 4(-1)

9. Ans.(A, B, C)	10. Ans.(A,D)	11. Ans.(A,C)	12. Ans.(A,B)
13. Ans. (C, (D)			

SECTION-II: (Integer Second Decimal) : 4(0)

1. Ans.(3.00)	2. Ans.(3.00)	3. Ans. (2.94 to 3.00)
4. Ans.(16.10 to 16.30)		

SECTION-IV : Matrix-Match Type : 8(0)

1. Ans.(A)-P,Q,R,S ; (B) P, Q, R, S, (C)- P, Q, R, S,T, (D)-P,Q

PHYSICAL CHEMISTRY

Advance : Guided Revision Plan - Question Bank- (02)

Topic : Chemical Equilibrium & Ionic Equilibrium

MM : 64
Time : 45 Min.

SECTION-I : (i) Only One option correct Type

This section contains **9 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **3(-1)**

- A buffer solution was prepared by dissolving 0.02 mole acetic acid & 0.01 mole sodium acetate in enough water to make 1.0 L of solution ($K_{a(\text{CH}_3\text{COOH})} = 2 \times 10^{-5}$). If 5×10^{-4} mole NaOH were added to 100 ml of the buffer. The resultant pH of the solution would be [given $\log 5 = 0.70$]
 (A) 5 (B) 4.7 (C) 5.3 (D) None
- PCl_5 (molecular mass = M) dissociates into PCl_3 and Cl_2 as : $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$. If the total pressure of the system at equilibrium is P and the density is 'd' at temperature, T K. The degree of dissociation of PCl_5 may be represented as
 (A) $\frac{PM}{dRT}$ (B) $\frac{PM}{dRT} - 1$ (C) $\frac{dRT}{PM} - 1$ (D) $\frac{dRT}{PM}$
- Solid Ammonium carbamate dissociates as: $\text{NH}_2\text{COONH}_4(\text{s}) \rightleftharpoons 2\text{NH}_3(\text{g}) + \text{CO}_2(\text{g})$. In a closed vessel solid ammonium carbamate is in equilibrium with its dissociation products. At equilibrium, ammonia is added such that the partial pressure of NH_3 at new equilibrium now equals the original total pressure. Calculate the ratio of total pressure at new equilibrium to that of original total pressure.
 (A) $\frac{31}{9}$ (B) $\frac{31}{27}$ (C) $\frac{27}{31}$ (D) $\frac{9}{31}$
- Which of the following can not act as buffer solution -
 (A) 30 ml 0.1M CH_3COOH + 10 ml 0.3M NH_4OH
 (B) 10 ml 0.2M Na_2CO_3 + 20 ml 0.2M H_2CO_3
 (C) 30 ml 0.1M HCl + 10 ml 0.1M Na_3PO_4
 (D) 10 ml 0.1M NH_4Cl + 5 ml 0.1M NaOH
- A certain gas 'A' polymerizes to a very small extent at a given temperature as : $n\text{A}(\text{g}) \rightleftharpoons \text{A}_n(\text{g})$. The reaction is started with one mole of 'A' in a container of capacity V. Which of the following is correct value of $\frac{PV}{RT}$, at equilibrium?
 (A) $1 - \frac{(n-1) \cdot K_c}{V^{n-1}}$ (B) $\frac{(n-1) \cdot K_c}{V^{n-1}}$ (C) $1 - \frac{n \cdot K_c}{V^{n-1}}$ (D) $1 - \frac{n \cdot K_c}{V^{n-1}}$
- The increasing order of Ag^+ ion concentration in :
 (I) Saturated solution of AgCl (II) Saturated solution of AgI
 (III) 1M $\text{Ag}(\text{NH}_3)_2^+$ in 0.1 M NH_3 (IV) 1M $\text{Ag}(\text{CN})_2^-$ in 0.1 M KCN
 Given : $K_{sp}(\text{AgCl}) = 10^{-10}$; $K_{sp}(\text{AgI}) = 10^{-16}$, $K_f(\text{Ag}(\text{NH}_3)_2^+) = 10^8$; $K_f(\text{Ag}(\text{CN})_2^-) = 10^{11}$
 (A) $\text{II} < \text{I} < \text{III} < \text{IV}$ (B) $\text{IV} < \text{III} < \text{II} < \text{I}$
 (C) $\text{IV} < \text{II} < \text{III} < \text{I}$ (D) $\text{IV} < \text{II} < \text{I} < \text{III}$

7. pH of a solution by mixing 50 mL of 0.1 M Na_3PO_4 and 50 mL of 0.05 M NaH_2PO_4 is :
 [For H_3PO_4 : $\text{pK}_1 = 2.1$, $\text{pK}_2 = 7.2$, $\text{pK}_3 = 12$ and $\log 3 = 0.3$]
 (A) 9.6 (B) 4.65 (C) 7.2 (D) 11.7
8. For the reaction
 $[\text{Ag}(\text{CN})_2]^- \rightleftharpoons \text{Ag}^+ + 2\text{CN}^-$; K_c at 25°C is 4×10^{-19}
 The equilibrium concentration of Ag^+ is, in a solution which is 0.1 M in KCN & 0.03 M in AgNO_3 originally.
 (A) $7.5 \times 10^{-16}\text{M}$ (B) $7.5 \times 10^{-18}\text{M}$ (C) $1.25 \times 10^{19}\text{M}$ (D) $1.25 \times 10^{-17}\text{M}$
9. The initial rate of hydrolysis of methyl acetate (1M) by a weak acid (HA, 1M) is $1/100^{\text{th}}$ of that of a strong acid (HX, 1M), at 25°C . The K_a of HA is
 (A) 1×10^{-4} (B) 1×10^{-5} (C) 1×10^{-6} (D) 1×10^{-3}

(ii) One or more options correct Type

This section contains **3 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. **4(-1)**

10. A reaction at 300K with $\Delta G^\circ = -1743\text{ J}$ consists of 3 moles of A(g), 6 moles of B(g) and 3 moles of C(g). If A, B and C are in equilibrium in 1 L container, then the reaction may be.
 ($\ln 2 = 0.7$, $R = 8.3\text{ J/K-mol}$).
 (A) $\text{A} + \text{B} \rightleftharpoons \text{C}$ (B) $\text{A} \rightleftharpoons \text{B} + 2\text{C}$ (C) $2\text{A} \rightleftharpoons \text{B} + \text{C}$ (D) $\text{A} + \text{B} \rightleftharpoons 2\text{C}$
11. Which of the following combination of solutes would result in the formation of a buffer solution ?
 (A) $\text{NH}_3 + \text{NH}_4\text{Cl}$ (B) $\text{CH}_3\text{COOH} + \text{NaOH}$ in 2 : 1 molar ratio
 (C) $\text{NH}_3 + \text{HCl}$ in 2 : 1 molar ration (D) $\text{CH}_3\text{COOH} + \text{NaOH}$ in 1 : 2 molar ratio
12. Choose the correct alternatives
 (A) K_{sp} of $\text{Fe}(\text{OH})_3$ in aqueous solution is 3.8×10^{-38} at 298 K. The concentration of Fe^{3+} will increase with decrease in H^+ ion concentration.
 (B) In a mixture of NH_4Cl and NH_4OH in water, a further amount of NH_4Cl is added. The pH of mixture will decrease.
 (C) An aqueous solution of each of the following salts NH_4I , $\text{Cr}(\text{NO}_3)_3$, KCN will be basic, acidic and neutral respectively.
 (D) Degree of hydrolysis of 0.1 M $\text{CH}_3\text{COONH}_4$, 0.2 M $\text{CH}_3\text{COONH}_4$, 0.5 M $\text{CH}_3\text{COONH}_4$ is same.

(iv) Table Type

- This section contains **3** questions of matching type.
- This section contains **ONE** tables (each having 3 columns and 4 rows)
- Based on each table, there are **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.

Zero Marks : 0 If none of the bubbles is darkened.

Negative Marks : -1 In all other cases

Column-I Compound	Column-II Relationship	Column-III Statement
(A) CH_3COOH ($K_a = 2 \times 10^{-5}$)	(P) $\text{pH} = \frac{1}{2} (\text{p}K_{a_1} + \text{p}K_{a_2})$	(I) Addition of HCl(aq) can make buffer solution
(B) 0.1 M NaHCO_3	(Q) $\text{pH} = \frac{1}{2} (\text{p}K_a - \log C)$	(II) Increases solubility of AgCl(s)
(C) $\text{NH}_3(\text{aq.})$ ($K_b = 2 \times 10^{-5}$)	(R) $\text{pH} = 7 + \frac{1}{2} (\text{p}K_b + \log C)$	(III) pH is independent of concentration
(D) $\text{CH}_3\text{COONa(aq.)}$	(S) $\text{pOH} = \frac{1}{2} (\text{p}K_b - \log C)$	(IV) Addition of NaOH (aq.) can make buffer solution

13. Which of the following option is incorrectly matched -

- (A) $A \rightarrow Q \rightarrow \text{IV}$ (B) $B \rightarrow S \rightarrow \text{IV}$ (C) $C \rightarrow S \rightarrow \text{II}$ (D) $C \rightarrow S \rightarrow \text{I}$

14. 30 ml of 0.1 M $\text{CH}_3\text{COOH (aq.)}$ is mixed with 10 ml of 0.1 M NaOH(aq.) pH of resulting solution

- (A) 4.7 (B) 5.3 (C) 4.4 (D) 5.6

15. For H_2CO_3 ($K_{a_1} = 10^{-7}$; $K_{a_2} = 10^{-11}$)

Number of OH^- in 1 litre of 0.1 M $\text{NaHCO}_3(\text{aq.})$ is -

- (A) $10^{-5} \times N_A$ (B) $10^{-9} \times N_A$ (C) $10^{-7} \times N_A$ (D) $10^{-11} \times N_A$

PHYSICAL CHEMISTRY

Score Advance : Guided Revision Plan - Question Bank- (03)

Topic : Thermodynamics & Thermochemistry

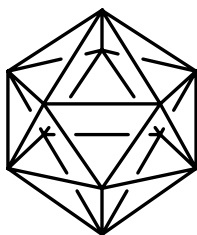
MM : 67

Time : 45 Min.

SECTION-I : (i) Only One option correct Type

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **3(-1)**

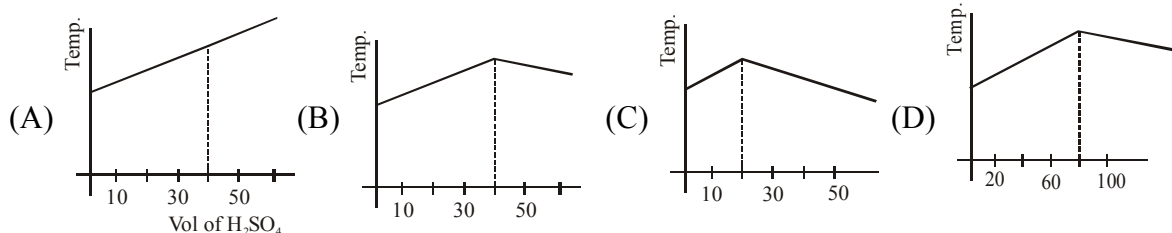
1. Boron exist in different allotropic forms. All allotropic form contains icosahedral units (icosahedral is a regular shape with 12 corners & 20 faces) with boron atoms at all 12 corners and all bonds are equivalent.



Icosahedral

Calculate $|\Delta H|$ (in kJ) per mole of boron atoms forming gaseous icosahedral if $\Delta H_{BE} (B-B) = 200$ kJ/mol.

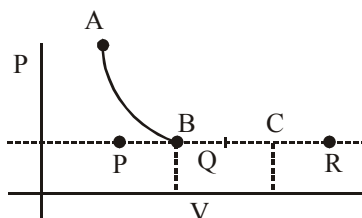
- (A) 1000 (B) 600 (C) 500 (D) 250
2. In an experiment the enthalpy of neutralisation of sodium hydroxide with sulphuric acid, 50cm^3 of 0.4M sodium hydroxide were titrated thermometrically with 0.25M sulphuric acid. Which of the following plots give the most probable correct representation temperature of solution (initial temperature of NaOH & H_2SO_4 are same).



3. 10 litre of an ideal gas at 25 atm & 27°C is expanded isothermally to 1 atm against a constant external pressure of 760 torr . Calculate work done by the gas in litre-atm
- (A) 240 (B) 250 (C) 800 (D) 400
4. Which of the following reactions is spontaneous only at relatively low temperature -
- (A) $\text{NH}_4\text{Br(s)} + 188\text{ kJ} \rightarrow \text{NH}_3\text{(g)} + \text{Br}_2\text{(l)}$
- (B) $\text{NH}_3\text{(g)} + \text{HCl(g)} \rightarrow \text{NH}_4\text{Cl(s)} + 176\text{ kJ}$
- (C) $2\text{H}_2\text{O}_2\text{(l)} \rightarrow 2\text{H}_2\text{O(l)} + \text{O}_2\text{(g)} + 196\text{ kJ}$
- (D) Both (B) & (C)

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5. For an ideal gas, three adiabatic processes are carried out upto same final pressure from same initial state. If adiabatic reversible process ends up at point 'B' and adiabatic single step irreversible process ends up at point 'C', then adiabatic free expansion upto same final pressure will end up at -



- (A) A (B) P (C) Q (D) R
6. Which of the options contain correct set of parameters which are both amount and path independent parameters.
- (I) Internal energy
 (II) Specific enthalpy
 (III) Heat exchange at constant pressure condition
 (IV) Molar heat exchange at constant volume condition
 (V) Temperature
 (VI) Work involved
- (A) Only II, IV & V (B) All option except VI
 (C) All options except III, IV & VI (D) Only II & V

(ii) One or more options correct Type

This section contains **5 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. **4(-1)**

7. Assume ideal gas behaviour for all the gases considered & vibrational degrees of freedom to be active. Separate equimolar samples of Ne, O₂, SO₂ & CH₄ were subjected to a two step process as mentioned. Initially all are at same state of temperature & pressure.
- Step-I** → All undergo reversible adiabatic expansion to attain same final volume, which is double the original volume thereby causing the decreases in their temperature.
- Step-II** → After step I all are given appropriate amount of heat isochorically to restore the original temperature.
- Mark the correct option(s)
- (A) Due to step I only, the decrease in temperature will be maximum for Ne.
 (B) During step II, heat given will be minimum for CH₄
 (C) There will be no net change in internal energy for any of the gas after both the steps of process are completed.
 (D) The P-V graph of CH₄ & SO₂ will be same for overall process.
8. Choose the correct statement(s)
- (A) During adiabatic expansion of an ideal gas magnitude of work obtained is equal to ΔH of gas.
 (B) For same change in temperature of ideal gas through adiabatic process, magnitude of W will be same in reversible as well as irreversible process
 (C) During boiling of H₂O at 1 atm, 100°C kinetic energy of H₂O molecules increases.
 (D) For same change in volume of ideal gas, ΔU is more in adiabatic expansion than in isobaric expansion if expansion starts with same initial state.

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9. Select the correct option(s):
- (A) $\Delta H_f^\circ [\text{H}_2\text{O}(l)]$ is zero at 25°C
- (B) $\Delta S_f^\circ [\text{O}_2(g)]$ is zero at 25°C
- (C) $S^\circ [\text{O}_2(g)]$ is zero at 25°C
- (D) Entropies of aqueous ions can be negative or positive
10. Select the **incorrect** statement (s) -
- (A) $\Delta H_{\text{atomisation}}$ of graphite is equal to $\Delta H_f [\text{C}(g)]$
- (B) ΔH_{comb} of 'H' atom is equal to $\Delta H_f [\text{H}_2\text{O}(l)]$
- (C) $\Delta H_f [\text{H}_2\text{O}(l)]$ is zero
- (D) ΔH_{comb} of graphite is equal $\Delta H_f [\text{CO}(g)]$
11. Which of the following is correct option(s)-
- (ϵ : Bond enthalpy, $\Delta_a H$: enthalpy of atomisation)
- (A) $(\Delta_f H)_{\text{H}(g)} = \frac{1}{2} (\Delta_a H)_{\text{H}_2(g)} = \frac{1}{2} \epsilon_{\text{H-H}}$
- (B) $(\Delta_f H)_{\text{C}(g)} = (\Delta_a H)_{\text{C}(gr,s)} = (\Delta_{\text{sub}} H)_{\text{C}(gr,s)}$
- (C) $(\Delta_f H)_{\text{Br}(g)} = \frac{1}{2} (\Delta_a H)_{\text{Br}_2(l)} = \frac{1}{2} \epsilon_{\text{Br-Br}}$
- (D) $(\Delta_f H)_{\text{I}(g)} = (\Delta_a H)_{\text{I}_2(s)} = (\Delta_{\text{sub}} H)_{\text{I}_2(s)}$

(iii) Paragraph Type

This section contains **1 paragraphs** each describing theory, experiment, data etc. **3 questions** relate to one paragraph with these questions. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D). **3(0)**

Paragraph for Q. No. 12 to 14

Study the following thermochemical equation :

- (a) $\text{N}_2\text{O}(g) + 3\text{H}_2(g) \rightarrow \text{H}_2\text{O}(l) + \text{N}_2\text{H}_4(l)$; $\Delta H = -76 \text{ kcal}$
- (b) $4\text{N}_2(g) + 3\text{H}_2\text{O}(l) \rightarrow 2\text{NH}_3(g) + 3\text{N}_2\text{O}(g)$; $\Delta H = +240 \text{ kcal}$
- (c) $\text{N}_2\text{H}_4(l) + \text{H}_2\text{O}(l) \rightarrow 2\text{NH}_3(g) + \frac{1}{2} \text{O}_2(g)$; $\Delta H = +36 \text{ kcal}$
- (d) $\text{H}_2\text{O}(l) \rightarrow \text{H}_2(g) + \frac{1}{2} \text{O}_2(g)$; $\Delta H = 68 \text{ kcal}$

12. The enthalpy of combustion (in kcal/mol) of liquid N_2H_4 is -
- (A) -129 (B) -138 (C) -147 (D) 147
13. The enthalpy of formation (in kcal / mol) of liquid N_2H_4 is -
- (A) -11 (B) +44 (C) +11 (D) +13
14. Ammonia can be obtained according to reaction (b) or (c). Which of the following is correct regarding these reactions, for the same mass of ammonia formed.
- (A) More mass of reactants is needed for reaction (c)
- (B) Less mass of reactants is needed for reaction (b)
- (C) More heat is needed for reaction (b)
- (D) More heat is needed for reaction (c)

SECTION-II: (Integer Second Decimal)

- This section contains 3 questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777.... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+		-	
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If ONLY the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- Pressure over 1000 mL of a liquid is gradually increases from 1 bar to 1001 bar under adiabatic conditions. If the final volume of the liquid is 990 mL, and ΔH of the process, assuming linear variation of volume with pressure, is 'x' kJ, then the value of (x) is.
- One mole of ideal gas undergoes following cyclic process:
 - Isochoric heating from (P_1, V_1, T_1) to double temperature.
 - Isobaric expansion from to double volume.
 - Linear expansion (on PV curve) to $(P_1, 8V_1)$.
 - Isobaric compression to initial state.

If the initial temperature of the gas is 300 K and the magnitude of work done (in kcal) by the gas.

$$\text{Given: } R = 2 \frac{\text{Cal}}{\text{mol-K}}$$

- $2A(s) \rightleftharpoons B(g) + 2C(g) + 3D(g)$

Total pressure developed in closed container by decomposition of A at equilibrium is 12 bar at 727°C. If ΔG° (kcal), of the reaction at 727°C is ($R = 2\text{cal/mole-K}$, $\ln 2 = 0.7$, $\ln 3 = 1.1$)

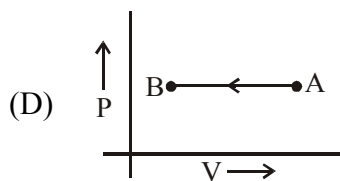
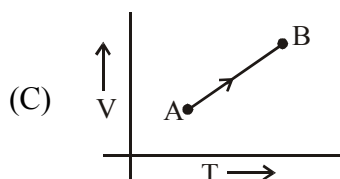
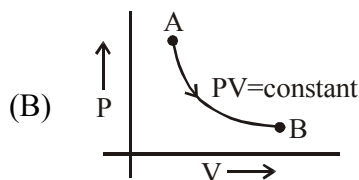
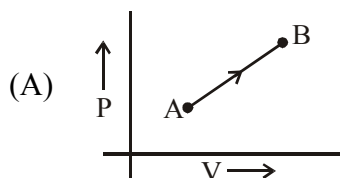
SECTION-IV : Matrix-Match Type

This Section contains **1 question**. Question has **four graphs** (A, B, C and D) given in **Column I** and five parameters (P, Q, R, S and T) in **Column II**. Any given graph in Column I can have correct matching with **ONE** or **MORE** parameters given in Column II. For example, if for a given question, statement B matches with the parameters given in Q and R, then for the particular question, against graph B, darken the bubbles corresponding to Q and R in the ORS.

8(0)

1. One mole of an ideal gas undergoing from state A to State-B through different processes column-I shows graph of the process & column-II shows change in parameter in these process. Match the following

Column-I



Column-II

(P) $q > 0$

(Q) $W > 0$

(R) $\Delta U > 0$

(S) $\Delta H > 0$

(T) $W < 0$

ANSWER KEY

SECTION-I : (i) Only One option correct Type: 3(-1)

- | | | |
|---------------|---------------|---------------|
| 1. Ans.(C) | 2. Ans.(B) | 3. Ans.(A) |
| 4. Ans.(B) | 5. Ans.(D) | 6. Ans.(A) |

(ii) One or more options correct Type : 4(-1)

- | | | |
|--------------------|------------------|------------------|
| 7. Ans.(A,C) | 8. Ans. (B) | 9. Ans. (B,D) |
| 10. Ans. (B,C,D) | 11. Ans. (A,B) | |

(iii) Paragraph Type: 3(0)

- | | | |
|---------------|---------------|---------------|
| 12. Ans.(C) | 13. Ans.(C) | 14. Ans.(C) |
|---------------|---------------|---------------|

SECTION-II: (Integer Second Decimal) : 4(0)

- | | | |
|-------------------|------------------|--------------------|
| 1. Ans.(99.50) | 2. Ans.(2.40) | 3. Ans. (17.80) |
|-------------------|------------------|--------------------|

SECTION-IV : Matrix-Match Type: 8(0)

1. Ans (A) - P,R,S,T; (B) - P, T; (C) - P, R, S, T; (D) -Q

PHYSICAL CHEMISTRY

Score Advance : Guided Revision Plan - Question Bank- (04)

Topic : Thermodynamics & Thermochemistry

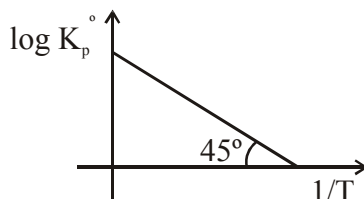
MM : 72

Time : 45 Min.

SECTION-I : (i) Only One option correct Type

This section contains **9 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **3(-1)**

- Each of the vessel 1 and 2 contain 1.2 moles of gaseous helium. The ratio of the vessel volumes is $V_2/V_1 = 2.0$, and the ratio of the absolute temperature of helium in them is $T_1/T_2 = 2.0$. Assuming the gas to be ideal, find the difference of gas entropies in these vessels, $S_2 - S_1$. ($\ln 2 = 0.7$).
 (A) 0.84 cal/ K (B) 4.2 cal/ K (C) -0.84 cal/ K (D) -4.2 cal/ K
- The entropy change accompanying the transfer of 12000 J of heat from a body A at 327°C to a body B at 127°C is -
 (A) -10.0 J/K (B) +10.0 J/K (C) -57.8 J/K (D) +57.8 J/K
- $2\text{NO}_2(\text{g}) \rightleftharpoons \text{N}_2\text{O}_4(\text{g})$
 $\Delta U_f^\circ [\text{N}_2\text{O}_4(\text{g})] = 2\text{kcal/ mole}$
 and $\Delta U_{\text{reaction}}^\circ = -16 \text{ kcal /mol}$ then calculate $\Delta H_{\text{formation}}^\circ$ of $\text{NO}_2 (\text{g})$ at 727°C
 (A) 9 kcal /mol (B) 4.5 kcal /mol (C) 8 kcal /mol (D) 10 kcal /mol
- For a process which follows the equation $PV^3 = C$, the work done when one mole of ideal gas was taken from 1 atm to $2\sqrt{2}$ atm starting from initial temperature of 300 K is
 (A) 300 R (B) 150 R (C) 600 R (D) 900 R
- For a reaction graph of $\log K_p^\circ$ and $1/T$ as follows -



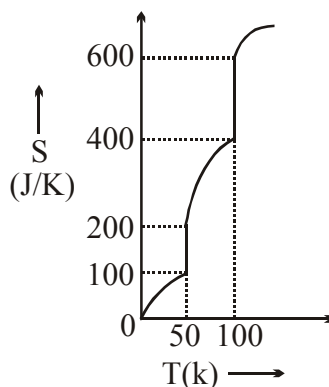
then standard enthalpy change for reaction will be -

- (A) 2 cal/mol (B) -2 cal/mol (C) 4.606 cal/mol (D) -4.606 cal/mol
- Which one of the following statement is false.
 (A) For a fix amount of a substance entropy of a substance in the liquid phase will be lower than the entropy of the same substance in gaseous phase
 (B) Spontaneous process always occur very rapidly
 (C) for a fix amount of a solid ΔS sublimation of a solid always positive
 (D) If $\Delta G < 0$, then it is not necessary that process will be always spontaneous
 - Heat liberated by a given amount of an ideal gas undergoing reversible isothermal process is 1200 cal at 300K. What is the Gibbs free energy change of the gas in this process ?

(A) zero (B) +1200 cal (C) -1200 cal (D) 4 cal

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8. ΔH_f^0 of water is $-285.8 \text{ kJ mol}^{-1}$. If enthalpy of neutralisation of monoacid strong base is $-57.3 \text{ kJ mol}^{-1}$, ΔH_f^0 of $\text{OH}^- (\text{aq.})$ ion will be
 (A) $-228.5 \text{ kJ mol}^{-1}$ (B) $228.5 \text{ kJ mol}^{-1}$ (C) $114.25 \text{ kJ mol}^{-1}$ (D) $-114.25 \text{ kJ mol}^{-1}$
9. For 1 mole substance variation of absolute entropy with temperature is given by following graph. What is **molar enthalpy of vapourisation** (ΔH_{vap}).



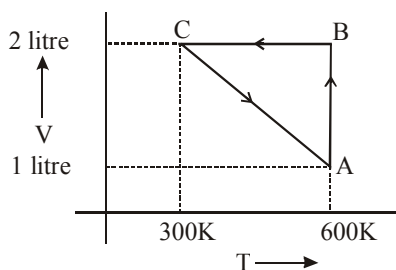
- (A) 2 kJ (B) 5 kJ (C) 20 kJ (D) 10 kJ

(ii) One or more options correct Type

This section contains **3 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. 4(-1)

10. Suppose that the volume of a certain ideal gas is to be doubled by one of the following processes
 (A) isothermal expansion (B) adiabatic expansion
 (C) free expansion in insulated condition (D) expansion at constant pressure.
- If E_1, E_2, E_3 and E_4 respectively are the changes in average kinetic energy of the molecules for the above four processes, then-
- (A) $E_2 = E_3$ (B) $E_1 = E_3$ (C) $E_1 > E_4$ (D) $E_4 > E_3$

11. 1 mole $\text{N}_2(\text{g})$ undergo following cyclic process -



which option (s) is/are correct - [Use : $R = 2 \text{ Cal / K - mole}$; $\ln 2 = 0.7$]

- (A) $W_{AB} = -840 \text{ cal}$ (B) $\Delta H_{AB} = 300 \text{ cal}$
 (C) $\Delta U_{CA} = 1500 \text{ cal}$ (D) $q_{BC} = -600 \text{ cal}$.
12. For $\text{MX}_2(\text{s})$ lattice enthalpy is 800 kJ mol^{-1} and enthalpy of solution is 50 kJ mol^{-1} . If hydration energies of M^{2+} and X^- ions are in the ratio 3 : 1, then correct options are :
- (A) Hydration energy of M^{2+} is -300 kJ/mol (B) Hydration energy of M^{2+} is -450 kJ/mol
 (C) Hydration energy of X^- is -300 kJ/mol (D) Hydration energy of X^- is -150 kJ/mol

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(iv) Table Type

- This section contains **1** question of matching type.
- This section contains **ONE** table (each having 3 columns and 4 rows)
- Based on each table, there are **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.

Zero Marks : 0 If none of the bubbles is darkened.

Negative Marks : -1 In all other cases

Column-I	Column-II	Column-III
(Reactions)	(Characteristics)	(Characteristics)
(I) $C(s) + O_2(g) \rightarrow CO_2(g)$	(i) $\Delta H = +ve$	(P) $\Delta H > \Delta E$
(II) $SO_3(g) \rightarrow SO_2(g) + \frac{1}{2} O_2(g)$	(ii) $\Delta n_g = +ve$	(Q) $ \Delta H > \Delta E $
(III) $2CO(g) + O_2(g) \rightarrow 2CO_2(g)$	(iii) $\Delta n_g = -ve$	(R) $\Delta H < \Delta E$
(IV) $C(s) + \frac{1}{2} O_2(g) \rightarrow CO(g)$	(iv) $\Delta H = -ve$	(S) $ \Delta H < \Delta E $
13. Which of the following is only correct match ?		
(A) I , i , P	(B) II , ii , Q	(C) III , iii , S
		(D) IV , iv , R
14. Which of the following is only incorrect match ?		
(A) II , i , P	(B) III , iv , R	(C) IV , iii , S
		(D) II , ii , Q
15. Which of the following is only correct match ?		
(A) I , ii , Q	(B) II , iii , Q	(C) III , iv , S
		(D) IV , ii , P

SECTION-II: (Integer Second Decimal)

- This section contains 4 questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places; e.g. 6.25, 7.00, -0.33, -30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

		+			
●	●	0	0	0	0
1	1	1	1	1	1
2	2	2	2	●	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	●
6	6	6	6	6	6
7	7	●	●	7	7
8	8	8	8	8	8
9	9	9	9	9	9

		-			
●	●	0	0	●	●
1	1	1	1	1	1
2	2	2	2	●	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	●	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If ONLY the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- At 500 kilobar pressure, density of diamond and graphite are 3 g/cc and 2 g/cc respectively, at certain temperature 'T'. Find the value $|\Delta H - \Delta U|$ (kJ/mole) for the conversion of 1 mole of graphite to 1 mole of diamond at temperature 'T' :
- A solution of 8.4 gm of a newly discovered compound (molar mass = 80000) in 25 ml of solution shows a temperature rise of 0.04°C for complete oxygenation. Each mole of compound binds 10 moles of oxygen. If heat capacity of solution is 4.2 J/K- ml then calculate amount of heat released per mole of oxygen bound in kJ
- The increase in Gibbs free energy (in kJ) of 13 g of ethanol (density = 0.78 g cm³), when the pressure is increased isothermally from 1 bar to 3001 bar, is
- The entropy change for the isothermal reversible expansion of an ideal gas at 300K from a particular state A to a state B is 20 J/K. Calculate the work (in kJ) obtained in the process.

SECTION-IV : Matrix-Match Type

This Section contains **1 question**. Question has **four statements** (A, B, C and D) given in **Column I** and five statements (P, Q, R, S and T) in **Column II**. Any given statement in Column I can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in Q and R, then for the particular question, against statement B, darken the bubbles corresponding to Q and R in the ORS. **8(0)**

1. Match the column

Column-I	Column-II
(A) Vaporisation of water at 100°C & 1 atm	(P) $\Delta S > 0$
(B) Isothermal reversible expansion of an ideal gas	(Q) $\Delta H = 0$
(C) Adiabatic free expansion of ideal gas	(R) $\Delta G < 0$
(D) Isobaric heating of an ideal gas	(S) $W = 0$
	(T) $q > 0$

ANSWER KEY
SECTION-I : (i) Only One option correct Type : 3(-1)

- | | | |
|-------------|------------|-------------|
| 1. Ans.(C) | 2. Ans.(B) | 3. Ans. (C) |
| 4. Ans.(B) | 5. Ans.(C) | 6. Ans. (B) |
| 7. Ans. (B) | 8. Ans.(A) | 9. Ans.(C) |

(ii) One or more options correct Type : 4(-1)

- | | | |
|----------------|----------------|----------------|
| 10. Ans. (B,D) | 11. Ans. (A,C) | 12. Ans (B, D) |
|----------------|----------------|----------------|

(iv) Table Type :

- | | | |
|-------------|-------------|-------------|
| 13. Ans.(B) | 14. Ans.(C) | 15. Ans.(D) |
|-------------|-------------|-------------|

SECTION-II: (Integer Second Decimal) : 4(0)

- | | | |
|----------------|---------------|---------------|
| 1. Ans. 100.00 | 2. Ans.(4.00) | 3. Ans.(5.00) |
| 4. Ans (6.00) | | |

SECTION-IV : Matrix-Match Type : 8(0)

1. Ans.(A)–P, T ; (B)–P,Q,R,T ; (C)–P,Q,R,S ; (D)–P,R,T

Yashpatil TG~ @bohring_bot